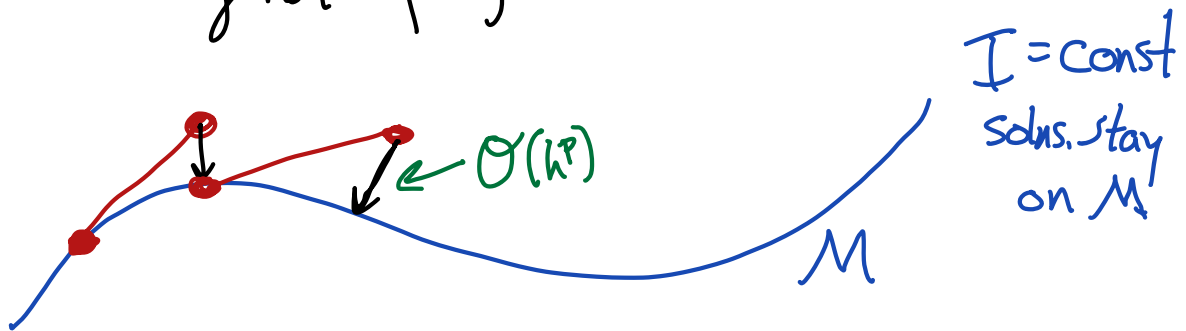


No RK method preserves all invariants.

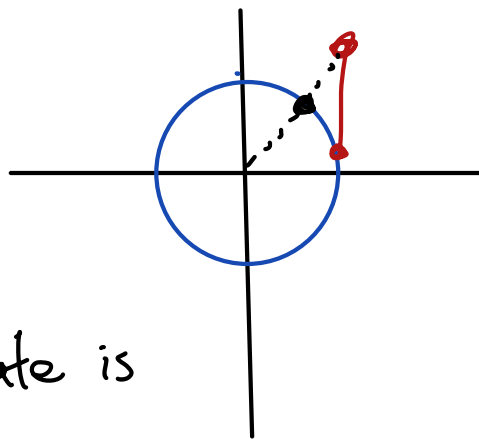
How to preserve general (nonlinear, non-quadratic) invariants?

① Orthogonal projections



How costly is this?

In general, we need to solve an algebraic eqn.



How accurate is this?

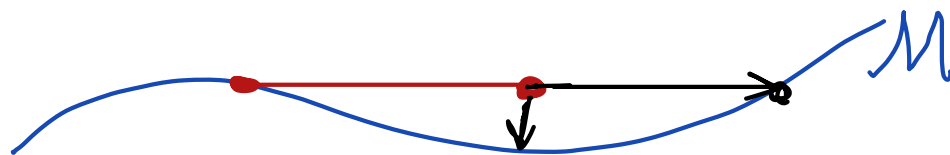
The order of accuracy is the same as that of the original method.

One drawback:

If we also have a linear invariant, it will be destroyed (in general).

Relaxation (Oblique projection)

$y_j: K$ stages



$$\|u_1\|^2 - \|u_0\|^2 = 2h \sum_{j=1}^s b_j y_j^T f(y_j) - 2k^2 \sum_{i,j=1}^s b_i a_{ij} f(y_i)^T f(y_j)$$

$$\text{If } \frac{d}{dt} \|u(t)\| = 0$$

$$+ k^2 \sum_i \sum_j b_i b_j f(y_i)^T f(y_j)$$

$$\text{then } u^T f(u) = 0$$

$$\text{Relaxation: } \hat{u}_1 = u_0 + h \gamma \sum_{j=1}^s b_j f(y_j)$$

$$\|\hat{u}_1\|^2 - \|u_0\|^2 = -2k^2 \gamma \sum_{i,j=1}^s b_i a_{ij} f(y_i)^T f(y_j)$$

$$+ k^2 \gamma^2 \sum_i \sum_j b_i b_j f(y_i)^T f(y_j) = 0$$

This always has one non-trivial root.

$$\gamma = \frac{2 \sum_{i,j=1}^s b_i a_{ij} f(y_i)^T f(y_j)}{\sum_i \sum_j b_i b_j f(y_i)^T f(y_j)}$$

The accuracy of $\hat{u}_1$: (for a method of order p)

$$\|\hat{u}_1 - u(h)\| = \mathcal{O}(h^p) \text{ even though } \|u_1 - u(h)\| = \mathcal{O}(h^{p+1})$$

But

$$\|\hat{u}_1 - u(\gamma h)\| = \mathcal{O}(h^{p+1})$$

So we view $\hat{u}_1$ as an approximation of u at $t = \gamma h$.

Unlike orthogonal projection, relaxation preserves linear invariants.

Homework for Sunday:

Choose a nonlinear problem with an invariant, and choose a numerical method. Compute the modified eqn. including the leading correction terms.

Use the modified equation to show (analytically) how the invariant will behave in the numerical solution.